

Classification metrics

Step – 1: Read the data

Step – 2: Preprocess the data

Step – 3: Model development

Step – 4: Model evaluation

Step – 5: Model tuning

Step – 6: Model deployment

Regression evaluation metrics

1) *RSE*

2) *MAE*

3) *MSE*

4) *RMSE*

5) *MAPE*

MSE VS MAE VS MAPE VS RMSE

Classification metrics

1) *Confusion Matrix*

2) *Accuracy*

3) *Error rate*

4) *TPR: True positive rate*

5) *TNR: True Negative Rate*

6) *Sensitivity*

7) *Specificity*

8) *Precision*

9) *Recall*

10) *F1 – score*

11) *Roc – AUC curve*

12) *cohen's kappa*

Confusion Matrix:

Assume that we are doing a classification problem, the target column has two labels

- yes
- No

when we pass the input model will give output

input data original output : actual output either yes or no

model output: predictions also either yes or no

How many combinations are available :

- Actually yes , Model predicted as yes : $Y - Y$
- Actually yes, Model predicted as No: $Y - N$
- Actualy No, Model predicted as Yes: $N - Y$
- Actually No, Model predicted as No: $N - N$

True Positives(TP):

Actually yes, Model predicted also yes

True Negtaives(TN):

Actually No, Model predicted also No

False Positives(FP):

Actually No, Model predicted also yes

False Negative(FN):

Actually yes, Model predicted also no

		Y	P	N
A	Y	TP	FN	
	N	FP	TN	

FP : There two letters

- *Consider the second letter : ' p '*
- *Second letter related to Model prediction*
- *In this case model predicted as yes*
- *First letter will tell , the prediction is correct or wrong*
- *F = the prediction wrong*
- *Actual value opposite to the prediction*
- *Model predicted as yes, actually No*

Accuracy:

Case – 1:

Actual	Model	
Yes	Yes	TP
No	Yes	FP
Yes	Yes	TP
No	No	TN
Yes	No	FN
No	No	TN

TP=2 TN= 2 FP =1 FN=1

Here out of total 6 observations 4 both are mathing

$$Tp = 2, Tn = 2$$

$$Accuracy = \frac{4}{6} = \frac{Tp + Tn}{No\ of\ observations} = \frac{Tp + Tn}{Tp + Tn + Fp + Fn} = \frac{2 + 2}{2 + 2 + 1 + 1}$$

Case – 2:

For example there are total 100 observations are there

Actaually yes: 60 ; Actually No: 40

Model predicted yes: 40, Model predicted No: 60

TP = 40 TN = 40 FP = 10 FN = 10(Wrong)

Note:

7 ===== Ay= 4 An=3

4 nos Model pred 3 yes mode

For example there are total 7 observations are there

Actaually yes: 4 ; Actually No: 3

Model predicted yes: 3, Model predicted No: 4

Actual	Model	
Yes	No	
Yes	No	
Yes	No	
Yes	No	
No	Yes	
No	Yes	
No	Yes	

TP, TN, FP, FN can calculate by Observation wise only

By seeing overall we will not able to identify

Case – 3:

Aditya is D – mart owner

Objective: He wants identify a terrorist while enter into the mass

He wants to use a ML model for thta

Aravind: Is a DS guy

Aravind: yes, I can do

Aravind: 100 people 99 are Not a terroist , 1 person is a terrorsit

Model behaviour:

It is predeicting correctly those who are Not a terrorsit: He is not a terrorsit

It is predicting wrong, those who actually a terrorist: He is not a terrorist

what is the accuracy:

Actual	Predicted
NT	NT
NT	

Data: Not a T: 99(No) T: 1 (yes)

Not a T =====> TN = 99

T =====> Not a T =====> FN = 1

TP = 0

Acc = 99%

Aravindh is veryhappy model accuracy : 99

Adithya also very happy : 99%

One day T :

Class imbalance problem:

- For any data set the classes means labels are not balanced
- In this case assume that : Yes cases are only one
 - No cases are 99
- Ratio : yes: No =====> 1: 99
- This type problems are called as class imbalance problem
- For imbalance problem, accuracy will fail
- Cancer patients
- Rare problems
 - Cancer problems
 - Objective: Identify the cancer : yes case
 - Generally Cancer patients are less
 - we want to identify yes, that are very very less
 - Model is not able to identify
 - Model become overfitted
- 100 === salman khan fan
- 1 === boby deol

How to overcome class imbalance problem:

- Under sampling(down sampling)
- Over sampling(up sampling)
- SMOTE(Synthetic Minority Over Sampling)

Whenever data has classes imbalanced accuracy will fail

TPR: True positive rate:

Out of total positive (yes) how many are exactly matching with model

$$TPR = \frac{TP}{P}$$

Case – 1:

Actual	Model	
Yes	Yes	TP
No	Yes	FP
Yes	Yes	TP
No	No	TN
Yes	No	FN
No	No	TN

$$TPR = \frac{TP}{P} = \frac{2}{3} = 66.6\%$$

Case – 2: Identify the Terrorist : yes

Originally No = 99 yes = 1

Predictions: All No cases predicting exactly No = TN

yes cases also predicting No = FN

TP = 0

$$TPR = \frac{0}{1} = 0\%$$

Acc = 99% ; TPR = 0%

What is TPR: out of total positive, how many my model is predicting

Aditya: what are positive case: Identify a terrorist

Aditya will not buy that model

TNR: True Negative rate:

Out of total negative (no) how many are exactly matching with model

$$TNR = \frac{TN}{N}$$

Case – 1:

Actual	Model	
Yes	Yes	TP
No	Yes	FP
Yes	Yes	TP
No	No	TN
Yes	No	FN
No	No	TN

$$TNR = \frac{TN}{N} = \frac{2}{3} = 66.6\%$$

Case – 2: Identify the Terrorist : yes

Originally No = 99 yes = 1

Predictions: All No cases predicting exactly No = TN

yes cases alsp predicting No = FN

TN = 99

$$TNR = \frac{99}{99} = 100\%$$

Sensitivity:

TPR (True positive rate also called as Sensitivity)

Specificity:

TNR (True Negative rate also called as Specificity)

$$TPR = \text{Sensitivity} = \frac{TP}{P}$$

$$TNR = \text{Specificity} = \frac{TN}{N}$$

Actual positives ===== Model predicted as positives
Actual negatives ===== Model predicted as Negatives

Case – 3

You developed one mode

Yes matching with model TP = 90

$$TN = 80$$

$$FP = 20$$

$$FN = 10$$

Accuracy ,TPR,TNR,Error rate

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} = \frac{90 + 80}{90 + 80 + 20 + 10} = \frac{170}{200} = 85\%$$

$$Error\ rate = 1 - Accuracy = 1 - 0.85 = 0.15 = 15\%$$

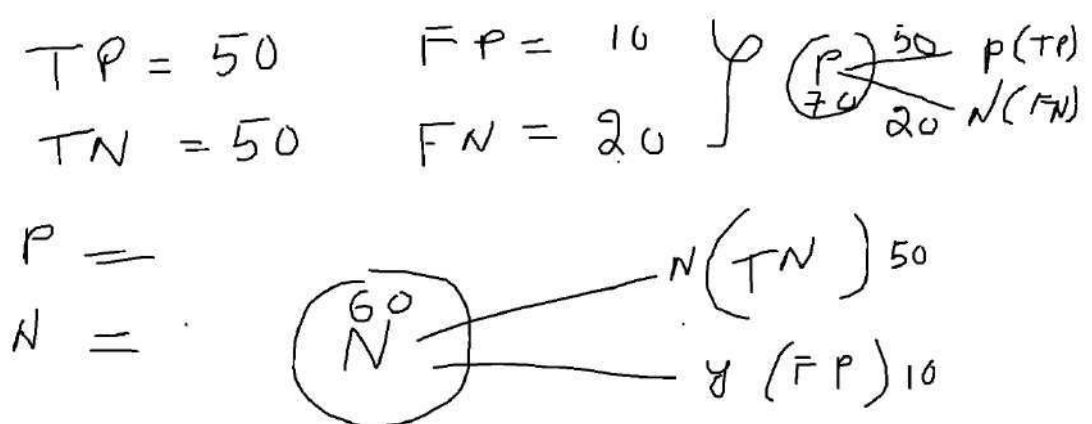
$$TPR = \frac{TP}{P} = \frac{90}{100} = 90\% =$$

$$P = TP + FN; \quad N = FP + TN$$

$$TNR = \frac{TN}{N} = \frac{80}{100} = 80\%$$

Accuracy = 85%,TPR = 90%,TNR = 80%

$$TP = 90, TN = 80, FP = 20, FN = 10$$



Accuracy = 85%, TPR = 90%, TNR = 80%

for example there are 10 actual positive cases are there, this machine easily will identify 9 cases

for example there are 10 actual negative cases are there, this machine easily identify 8 cases

Accuracy = 85%, TPR = 99%, TNR = 20%

for example there are 10 actual positive cases are there, this machine easily will identify almost all cases

for example there are 10 actual negative cases are there, this machine only identify 2 cases

TPR very helpful, if your goal has predicted positive cases

Accuracy = 85%, TPR = 20%, TNR = 99%

for example there are 10 actual positive cases are there, this machine only identify 2 cases

for example there are 10 actual negative cases are there, this machine identify cases

TNR very helpful, if your goal has predicted negative cases

Multiclassification:

Actual covid zones: Red, Green, Yellow

$TPR_{Red}, TPR_{Green}, TPR_{Yellow}$

TPR_{red} = out of actual red cases, how many model is predicting red

TPR_{green} = out of actual green cases, how many model is predicting green

TPR_{yellow} = out of actual yellow cases, how many model is predicting yellow

$$TPR_{avg} = \frac{TPR_{red} + TPR_{green} + TPR_{yellow}}{3}$$

$TNR_{Red}, TNR_{Green}, TNR_{Yellow}$

TNR_{red} = out of actual not red cases, how many model is predicting not red

TNR_{green} = out of actual not green cases, how many model is predicting not green

TNR_{yellow} = out of actual not yellow cases, how many model is predicting not yellow

$$TNR_{avg} = \frac{TNR_{red} + TNR_{green} + TNR_{yellow}}{3}$$

	RED	GREEN	YELLOW	
RED	TP_RED 10	20	50	80
GREEN	20	TP_GREEN 30	40	90
YELLOW	70	100	TP_YELLOW 120	290

Precision

		Y	P	N
A	Y	TP	FN	
	N	FP	TN	

$$Precision = \frac{TP}{TP+FP}$$

$$TP = 90, TN = 80, FP = 20, FN = 10$$

$$P = TP + FN, N = TN + FP$$

Total positive prediction made by model : TP+FP

no. of positive predictions by model which is actually true

portion of positive prediction out of total positive

Out of total positive predictions considering only true positives

$$Precision = \frac{TP}{TP + FP}$$

**Out of Total positives(TP + FP)predicted by model,
how many are matching with actual positives (TP)**

Model Actual

Model predicted some positives , in that positive how many exactly matching with actual positives

Precision = 90%

Model predicted 100 positives ===== Actual = 90

you written 20qns === 20qns === > 10

$$\frac{10}{20} = precision = 50\%$$

Recall:

$$Recall = \frac{TP}{TP + FN}$$

Out of total actual positives(TP + FN), how many exactly predicted(TP) by model

FP : Model predicted as yes, but actually No

There is No road === Model says road

FN: model predicted as No, but actually yes

Naresh IT at ameerpet , H. tech city

Tagore : Fake reviews

The mobile number showed is not working properly

Precision or Recall

- ***Precision and recall both are working on True positives only***
 - ***Precision related to FP type of Problem***
 - ***Less FP then precision will increase***
 - ***A patient dont have diagnosis, but model says yes***
 - ***The drug is not working, but model says yes***
 - ***The students not passed the exam, but model says yes***
- ***Recall related to FN type of Problem***
- ***Less FN then recall will increase***
- ***A patient has diagnosis, but model says no***
- ***The drug is working, but model says no***
- ***The students passed the exam, but model says no***

Based on Problem statement both are important

we need to reduce both FP and FN

F1 – score:

- ***F1 – score is a Trade off between Precision and Recall***
- ***Some times we are not able to take the decision of model performance based on Precision and Recall***
- ***We can get only one metric by combination of Precision and Recall***
- ***DT pr = 80% Rc = 82%***
- ***KNN pr = 85% rc = 83%***
- ***F1 score is the best metric to choose the better model***

$$F1 = \frac{2 * Precision * recall}{Precision + Recall}$$

$$AM = \frac{a+b}{2}$$

$$GM = \sqrt{ab}$$

$$HM = \frac{2}{\frac{1}{a} + \frac{1}{b}} = \frac{2ab}{a+b}$$

F1 score is harmonic progression of Precision and Recall

Why HM, why not AM?

AM: Average always affect by outliers

For extreme values it will affect

Precision = 0.9, Recall = 0.1

$$AM = \frac{0.9 + 0.1}{2} = 0.5$$

$$HM = \frac{2 * 0.9 * 0.1}{0.9 + 0.1} = 0.18$$

Precision = 0.9, Recall = 0.8

$$AM = \frac{0.9 + 0.8}{2} = 0.85$$

$$HM = \frac{2 * 0.9 * 0.8}{0.9 + 0.8} = 0.84$$

ROC – AUC:

ROC: Receiver operative Characteristics

AUC: Area under the curve

It is a curve or graph between Sensitivity and (1 – specificity)

$$\text{Sensitivity} = \text{TPR} = \frac{TP}{P} = \frac{TP}{TP + FN} = \text{Recall}$$

$$\text{specificity} = \text{TNR}$$

$$(1 - \text{specificity}): 1 - \text{TNR}$$

$$1 - \frac{TN}{N} = 1 - \frac{TN}{TN + FP} = \frac{TN + FP - TN}{TN + FP} = \frac{FP}{TN + FP} = \frac{FP}{N} = \text{False positive rate}$$

TPR vs FPR

Graph between True positive Rate (TPR) $\left(\frac{TP}{P}\right)$ vs False positive Rate (FPR) $\left(\frac{FP}{N}\right)$

FPR: X – axis ((1 – specificity))

TPR : Y – axis (Sensitivity)

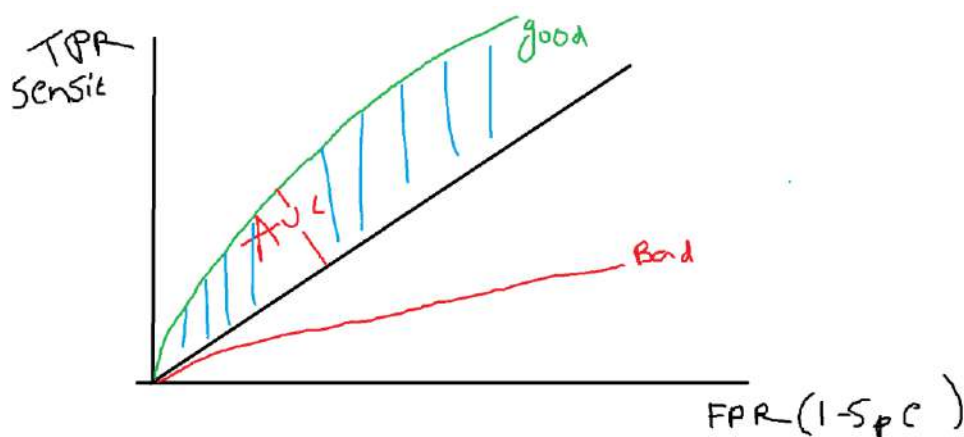
ROC – AUC curve also related to TP only

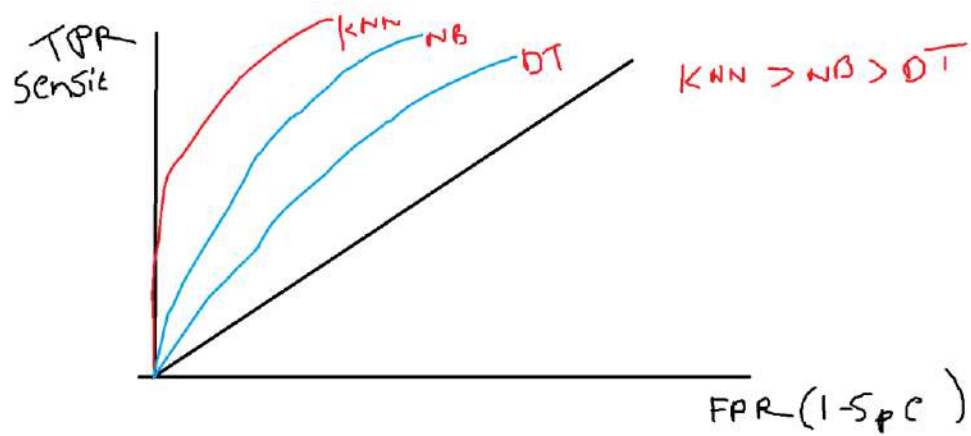
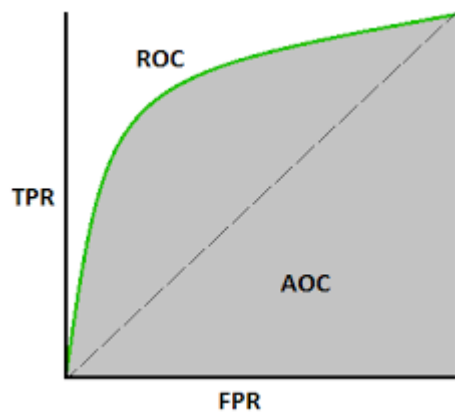
The more TPR, then the model is good

$$\text{TPR} = \frac{TP}{P} = \frac{TP}{TP + FN} = \text{Recall}$$

$$\text{Sensitivity} = \text{TPR} = \text{RECALL} = \frac{TP}{TP + FN}$$

$$\text{FPR} = \frac{FP}{N} = \frac{FP}{FP + TN}$$





Tuple	CLASS	Probability	Prediction	TP	FP	TP/P=TP/5	FP/N=FP/5	(FPR,TPR)
1	P	0.90	P	1	0	1/5=0.2	0	(0.2,0)
2	P	0.80	P	2	0			
3	N	0.70	P	2	1			
4	P	0.60	P					
5	P	0.55	P					
6	N	0.54	P					
7	N	0.53	P					
8	N	0.51	P					
9	P	0.50	P					
10	N	0.40	P					

Tuple	CLASS	Probability	Prediction	TP	FP	TP/P	FP/N	(FPR,TPR)
1	P	0.90	P	1	0			
2	P	0.80	P	2	0			
3	N	0.70	P	2	1			
4	P	0.60	P	3	1			
5	P	0.55	P	4	1			
6	N	0.54	P	4	2			
7	N	0.53	P	4	3			
8	N	0.51	P	4	4			
9	P	0.50	P	5	4			
10	N	0.40	P	5	5			

Tuple	CLASS	Probability	Prediction	TP	FP	FP/N	TP/P	
1	P	0.90	P	1	0	0	1/5=0.20	(0,0.2)
2	P	0.80	P	2	0	0	2/5=0.4	(0,0.4)
3	N	0.70	P	2	1	1/5=0.2	2/5=0.4	(0.2,0.4)
4	P	0.60	P	3	1	1/5=0.2	3/5=0.6	(0.2,0.6)
5	P	0.55	P	4	1	1/5=0.2	4/5=0.8	(0.2,0.8)
6	N	0.54	P	4	2	2/5=0.4	4/5=0.8	(0.4,0.8)
7	N	0.53	P	4	3	3/5=0.6	4/5=0.8	(0.6,0.8)
8	N	0.51	P	4	4	4/5=0.8	4/5=0.8	(0.8,0.8)
9	P	0.50	P	5	4	4/5=0.8	5/5=1	(0.8,1)
10	N	0.40	P	5	5	5/5=1	5/5=1	(1,1)

Case-1: Threshold 0.90 we are keeping

Probability of any observation ≥ 0.90 the answer is P otherwise False

First Observation prediction is P remaining all are No

[P,N,N,N,N,N,N,N,N]

Prediction always observation wise observation

Total actual Positives how many =5 (P)

Total actual Negatives how many =5 (N)

$$AUC: \Sigma \frac{(x_2 - x_1)(y_2 + y_1)}{2}$$

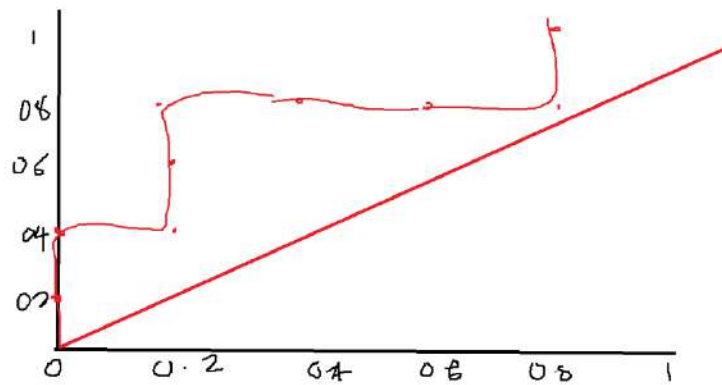
(0,0.2)
(0,0.4)

(0,0.4)
(0.2,0.4)

(0.2,0.4)
(0.2,0.6)

(0.2,0.6)
(0.2,0.8)

$$\begin{aligned} & \frac{(0 - 0)(0.4 + 0.2)}{2} + \frac{(0.2 - 0)(0.4 + 0.4)}{2} + \frac{(0.2 - 0.2)(0.6 + 0.4)}{2} + \frac{(0.2 - 0.2)(0.8 + 0.6)}{2} \\ & + \frac{(0.4 - 0.2)(0.8 + 0.8)}{2} + \frac{(0.6 - 0.4)(0.8 + 0.8)}{2} + \frac{(0.8 - 0.6)(0.8 + 0.8)}{2} \\ & + \frac{(0.8 - 0.8)(1 + 0.8)}{2} + \frac{(1 - 0.8)(1 + 1)}{2} = 0.76 \end{aligned}$$

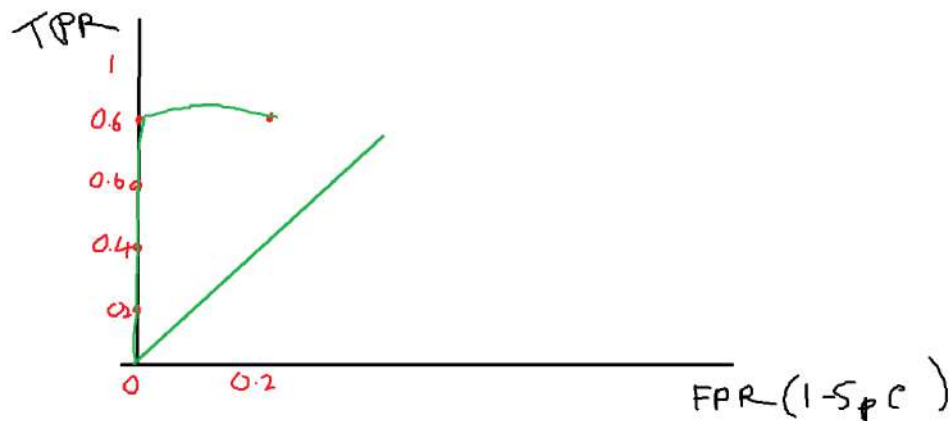


Total number of actual positives (P): 5

Total number of actual positives (N): 5

Probability means the confidence about the prediction = which is equal to actual

> 0.5 , The actual value only Transferred as Prediction



$AUC = 1$ Perfect model (Overfitting)

$AUC = 0.5$: Average

$AUC : 0.5 < AUC < 1$: Good model (Normal fit)

$AUC < 0.5$: Under fit

FPR more means what?

$$FPR \uparrow = \frac{FP \uparrow}{FP + \textcircled{TN} \downarrow}$$

$$TPR \downarrow = \frac{TP \downarrow}{TP + \textcircled{FN} \uparrow}$$

$$FP \Leftarrow \Rightarrow TN$$

$$FN \uparrow \Rightarrow TP \downarrow$$

Cohen's kappa

The agreement and Disagreement happens is by chance ?

main goal: TP and TN this should be always maximum

	Yes	NO
YES	a	b
NO	c	d

	Doctor B: D	Doctor A: H
Doctor A: D	40	10
Doctor B: H	20	30

$$k = \frac{p_o - p_e}{1 - p_e}$$

$p_o = \text{Observed Agreement}$

$p_e = \text{Agreement by Chance}$

Step – 1: Calculate the agreement

accuracy

$$p_o = \frac{\text{no of agreements}}{\text{total observations}} = \frac{40 + 30}{100} = \frac{70}{100} = 0.7$$

Step – 2: Calculate the agreement by chance

Doctor A told 50 members has Disease out of 100 members

Doctor B told 60 members has Disease out of 100 members

$$\text{Both doctors agreement by chance on Disease} = \frac{50}{100} * \frac{60}{100} = 0.3$$

Doctor A told 50 members has Healthy out of 100 members

Doctor B told 40 members has Healthy out of 100 members

$$\text{Both doctors agreement by chance on Healthy} = \frac{50}{100} * \frac{40}{100} = 0.2$$

$$\begin{aligned} \text{Total agreement} &= \text{Healthy}_{\text{Agreement}} + \text{Disease}_{\text{Agreement}} = 0.3 + 0.2 = 0.5 \\ &= p_e \end{aligned}$$

$$p_e = 0.5; \quad p_0 = 0.7$$

$$k = \frac{0.7 - 0.5}{1 - 0.5} = \frac{0.2}{0.5} = 0.4$$

- The Kappa statistic varies from 0 to 1, where.
- 0 = agreement equivalent to chance.
- 0.1 – 0.20 = slight agreement.
- 0.21 – 0.40 = fair agreement.
- 0.41 – 0.60 = moderate agreement.
- 0.61 – 0.80 = substantial agreement.
- 0.81 – 0.99 = near perfect agreement
- 1 = perfect agreement